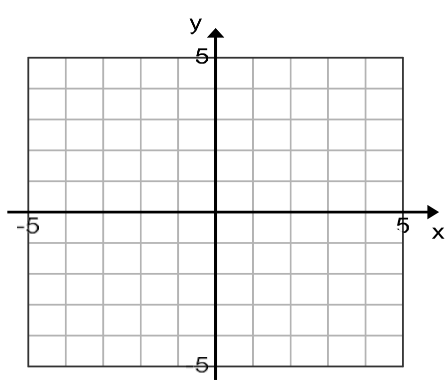
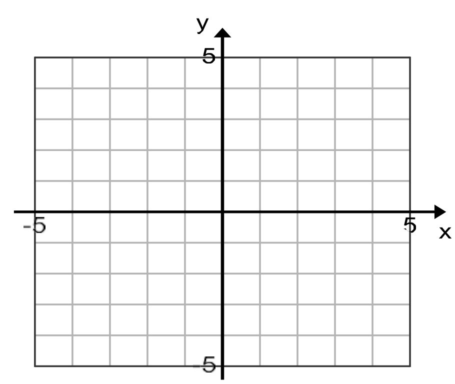
HOMEWORK DAY 12,13– *Polynomial Function (Section 7.2)*

1. Do as we did in **Example 1 Lecture 2** for each function. (Find intercepts, transformation, graph then domain and range)

(a) $f(x) = -\frac{1}{4}(x+1)^3 + 2$

(b) $p(x) = 3(x-2)^4 - 1$





2. Find the multiplicity (zeros of function) by factoring for each function below.

(a) $h(x) = -6x^3 - 9x^2 + 60x$

(b) $m(x) = x^5 - 10x^4 + 25x^3$

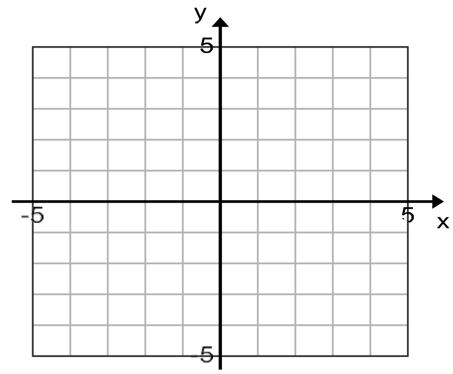
(c) $f(x) = 4x^4 - 37x^2 + 9$

(d) $n(x) = x^6 - 7x^4$

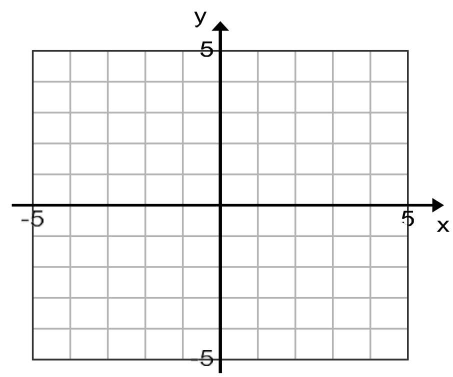
(e) $g(x) = x^5 - 2x^4$

3. Do as we do in **Example 3 Lecture 2**, find all zeros and graph.

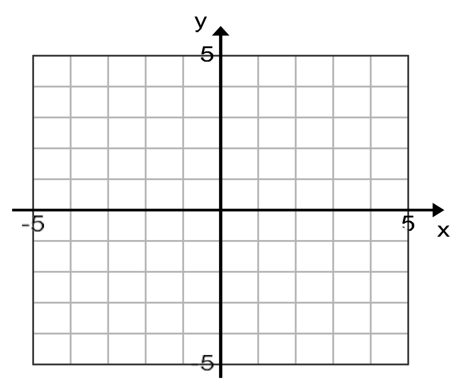
(a) $f(x) = x^2(x+4)(x-2)$



(b) $g(x) = x^5 - 2x^4$



(c) $h(x) = x^4 - 10x^2 + 9$



4. Find intercepts and asymptotes and delete points that exist and graph. State domain and range.

(a) $p(x) = \frac{4x}{x^3-6x^2+9x}$

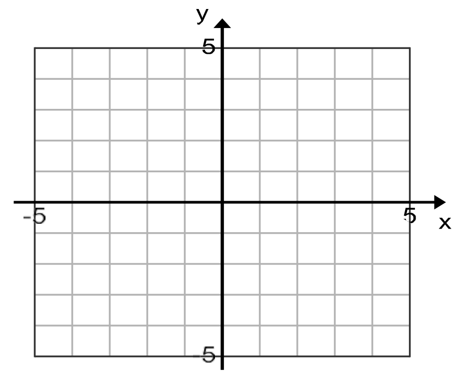
x-int:

VA:

y-int:

HA:

DP:



(b) $g(x) = \frac{x^3-3x^2-4x}{x+1}$

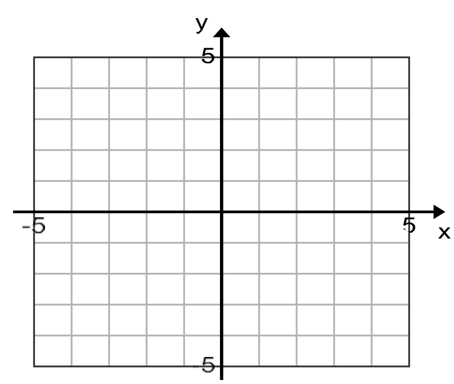
x-int:

VA:

y-int:

HA:

DP:



5. Find intercepts and asymptotes and delete points that exist.

(a) §8.2 16

x-int:
VA:
y-int:
HA:
DP:

(b) §8.2 18

x-int:
VA:
y-int:
HA:
DP:

(c) §8.2 20

x-int:
VA:
y-int:
HA: None
SA:
DP:

6. For each, find all intercepts, asymptotes & deleted points that exist, and graph. You must (1) completely FACTOR the function 1st and (2) justify why you have the function above/below the *x*-axis over each interval.

(a) $f(x) = \frac{2x+6}{3x-5}$

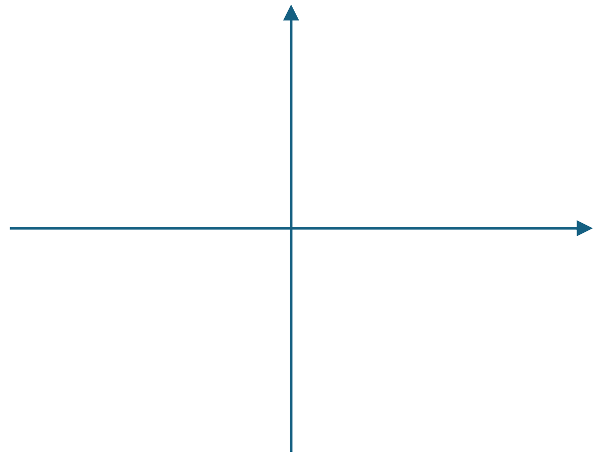
VA(s): ______________

x-int: _______________

HA(s): ______________

y-int: _______________

DP: ________________



x-intervals:

Above or Below:

Justification:

(b) $g(x) = \frac{x^2-2x+1}{x^2-4x+4}$

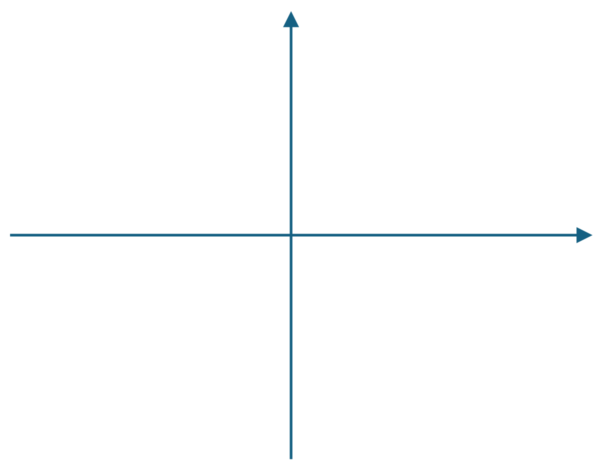
VA(s): ______________

x-int: _______________

HA(s): ______________

y-int: _______________

DP: ________________



x-intervals:

Above or Below:

Justification:

(c) $k(x) = \frac{x(x-1)^2}{x^3-3x^2-4x}$

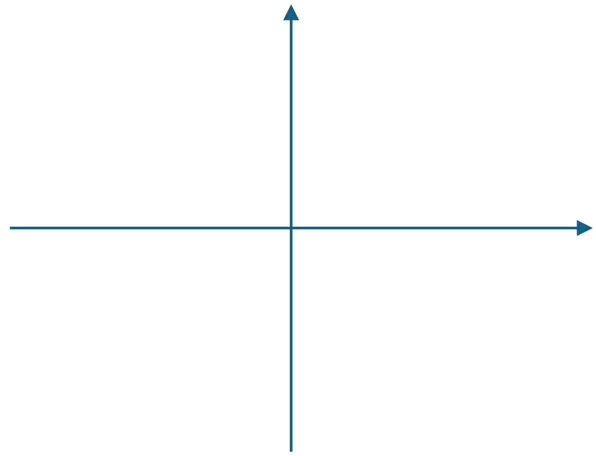
VA(s): _______________

x-int: ________________

HA(s): _______________

y-int: ________________

DP: _________________



x-intervals:	
Above or Below:	
Justification:	